The galactic acceleration scale as the Jeans rate of the vacuum

Mike One

Independent researcher, Italy

Corresponding author: Mike One, thegreatmemeresetlobby@gmail.com

Abstract

The acceleration scale of the radial acceleration relation, $g_\dagger \approx 1.2\times10^{-10}$ m s^{-2}, lies close to $cH_0/2\pi$, a coincidence without an accepted explanation. We propose that $g_\dagger$ is the Jeans rate of the vacuum energy density, $g_\dagger = c\sqrt{4\pi G\rho_\Lambda}/2\pi = c^2\sqrt{\Lambda/2}/2\pi = 1.056\times10^{-10}$ m s^{-2}, which is constant in time and agrees with $cH_0/2\pi$ today only because the vacuum dominates the present expansion. Taking the relaxed vacuum to be a mode in its ground state, thermally excited by a bath at the vacuum's Jeans temperature, and requiring scale invariance far from matter, yields the observed relation $g = g_N/\left(1-e^{-\sqrt{g_N/g_\dagger}}\right)$ with no free parameter. With pass/fail rules fixed before each run and no parameter fitted, this scale describes 153 SPARC rotation curves, passes two weak-lensing datasets (galaxy–galaxy circular velocities and the KiDS-1000 radial acceleration relation of late-type and blue lenses) and is consistent with 145 galaxies at $z = 0.7$–2.5, where growth with $H(z)$ is excluded at 3.4σ. The commonly quoted form $c^2\sqrt{\Lambda/3}/2\pi$ fails the three low-redshift datasets. The proposal predicts a constant scale for late-type lenses at $z \gtrsim 1$ and no dependence on the Hubble-constant tension. *Keywords:* radial acceleration relation; MOND acceleration scale; cosmological constant; vacuum energy; weak lensing

1 Introduction

Rotationally supported galaxies follow a tight radial acceleration relation (RAR) between the observed acceleration g and the Newtonian acceleration g_N of their baryons [1],

$$g = \frac{g_N}{1-e^{-\sqrt{g_N/g_\dagger}}}, \tag{1}$$

with a single acceleration scale $g_\dagger$. That this scale lies close to $cH_0/2\pi$ was noted when it was first measured [2], and it has since been related to the cosmological constant [3]. No derivation fixes both the functional form of Eq. (1) and the value of $g_\dagger$. This letter proposes one, and tests it.

2 The scale: the Jeans rate of the vacuum

A self-gravitating medium of density ρ has a natural gravitational rate, the Jeans frequency $\omega_J = \sqrt{4\pi G\rho}$. Treating the vacuum as a physical medium of density $\rho_\Lambda = \Lambda c^2/8\pi G$ gives

$$\omega_J = \sqrt{4\pi G\rho_\Lambda} = c\sqrt{\Lambda/2}, \qquad g_\dagger = \frac{c\,\omega_J}{2\pi} = \frac{c\sqrt{4\pi G\rho_\Lambda}}{2\pi} = \frac{c^2\sqrt{\Lambda/2}}{2\pi}. \tag{2}$$

With Planck 2018 parameters [4] ($H_0 = 67.4$ km s^{-1} Mpc^{-1}, $\Omega_\Lambda = 0.685$) this gives $g_\dagger = 1.056\times10^{-10}$ m s^{-2}. Since $c^2\Lambda/2 = \frac{3}{2}\Omega_\Lambda H_0^2$, Eq. (2) equals $cH_0/2\pi$ to within 1.4% at the present epoch, because $\Omega_\Lambda \simeq 2/3$ today. In the past $H(z)$ was larger, but ρ_Λ was not, so the scale of Eq. (2) stays constant while $cH(z)/2\pi$ grows.

The physical reading is a comparison of rates. Matter relaxes the vacuum where its own gravitational rate, g_N/c, exceeds the vacuum's, and the transition lies where the two are comparable.

3 The form: ground-state occupation and scale invariance

Postulate. The relaxed vacuum around a mass is a vacuum mode in its ground state. The unrelaxed vacuum is the same mode thermally excited by a bath set by the vacuum itself, at temperature $kT = \hbar\omega_J/2\pi$, which has the form of the Gibbons–Hawking temperature [5] with $H \to \omega_J$. The mass sets the mode.

Statistics. For a bosonic mode with $x = E/kT$, the ground-state probability is $1 - e^{-x}$ and the mean occupation is $\langle n\rangle = 1/(e^x - 1)$. Identifying the relaxed fraction with the ground-state probability, and the gravitational boost with $\nu = 1/(\text{relaxed fraction})$, gives

$$\nu = \frac{1}{1 - e^{-x}} = 1 + \langle n\rangle. \tag{3}$$

Scale invariance. Far from matter the unrelaxed vacuum is taken to be invariant under a common rescaling of time and space, the symmetry known to underlie the deep-MOND regime [6]. Write $x = (g_N/g_\dagger)^p$. For $x \ll 1$, $\nu \simeq 1/x$ and $g \simeq g_N^{1-p} g_\dagger^p$. Under the rescaling by λ, $g_N \to g_N/\lambda^2$, while a scale-invariant acceleration must scale as $g \to g/\lambda$. This fixes $2(1-p) = 1$, i.e. $p = \frac{1}{2}$. Equivalently, the mode energy is the geometric mean of the matter rate and the bath frequency. Then

$$g = \nu\, g_N = \frac{g_N}{1 - e^{-\sqrt{g_N/g_\dagger}}}, \tag{4}$$

which is Eq. (1), with $g_\dagger$ given by Eq. (2) and no free number. In the deep regime it gives $g = \sqrt{g_N g_\dagger}$, flat rotation curves and $v^4 = GMg_\dagger$.

4 Tests

All tests compare Eq. (1) with $g_\dagger$ fixed by Eq. (2) against two alternatives: the expansion form $cH(z)/2\pi$, and the commonly quoted vacuum form $c^2\sqrt{\Lambda/3}/2\pi = 0.863 \times 10^{-10}$ m s^{-2}. Pass rules were fixed before running, and no parameter was fitted to the data being tested.

1. **SPARC** [7]: 153 rotation curves (3135 points), with stellar mass-to-light ratio, distance and inclination marginalised per galaxy. Rule: $\Delta\text{BIC} < 10$ relative to the same law with $g_\dagger$ free.
2. **Galaxy–galaxy weak lensing**: circular velocities of isolated lenses at $z \approx 0.24$–0.39 [8], 20 points. Rule: $\Delta\chi^2 < 4$ from the best single $g_\dagger$.
3. **KiDS-1000 lensing RAR** [9], late-type ($n < 2.5$) and blue lenses only, to avoid the early-type excess, 30 points at $z \approx 0.25$. Same rule.
4. **High-redshift kinematics**: 145 discs at $z = 0.7$–2.5 from Price et al. [10], KROSS [11], KMOS3D [12] and KGES [13]. The statistic $R = \ln[g_\dagger(z)/g_\dagger^{\text{local}}]/\ln[H(z)/H_0]$ is 0 for a constant scale and 1 for the expansion form.

The Jeans scale passes all four tests. The expansion form fails the high-redshift test, and the $\sqrt{\Lambda/3}$ form fails the three low-redshift tests.

5 Discussion

Relation to earlier work. Milgrom [3] related MOND inertia to the Unruh temperature of an accelerated observer in de Sitter space, using temperature differences. This gives $a_0 \propto \sqrt{\Lambda/3}$, with interpolating functions that differ from Eq. (1). The present proposal differs in three ways: it uses occupation statistics rather than temperature differences, it derives the observed functional form, and it sets the factor through the vacuum's Jeans rate, $\sqrt{\Lambda/2}$, which the data prefer over $\sqrt{\Lambda/3}$.

Limitations.

Dataset	Jeans rate, Eq. (2)	$cH(z)/2\pi$	$c^2\sqrt{\Lambda/3}/2\pi$
SPARC (ΔBIC; free fit 1.054×10^{-10})	-7.8	-7.8	$+53$
Galaxy–galaxy lensing ($\Delta\chi^2$; best $1.21^{+0.19}_{-0.17}\times10^{-10}$, 2σ)	$+3.2$	$+0.9$	$+19.0$
KiDS late/blue lensing ($\Delta\chi^2$; best $1.15^{+0.19}_{-0.16}\times10^{-10}$, 2σ)	$+1.2$	$+0.2$	$+13.8$
High-z kinematics ($R=+0.33\pm0.20$)	1.7σ	excluded, 3.4σ	1.7σ

Table 1: Results. All three forms are fixed in advance. Pass: $\Delta\mathrm{BIC}<10$, $\Delta\chi^2<4$, $|R-R_{\mathrm{pred}}|<2\sigma$.

- At $z<0.5$, Eq. (2) and $cH(z)/2\pi$ differ by about 0.05 dex, so the low-redshift tests mainly confirm $g_\dagger\simeq cH_0/2\pi$ and reject $\sqrt{\Lambda/3}$. The separation from the expansion form rests on the high-redshift test, whose four samples disagree with one another ($\chi^2=9.9$ for 3 degrees of freedom).
- The rejection of $\sqrt{\Lambda/3}$ depends on stellar masses, whose systematic uncertainty is comparable to the 20% difference involved.
- The two lensing datasets share KiDS lenses.
- Early-type lenses show excess lensing signal in every model, which is not addressed here.
- The ground-state postulate was formulated with Eq. (1) in view. Its support lies in the value and constancy of the scale, which were not fitted.

Predictions.

1. The scale is constant for late-type lenses at $z\gtrsim1$, where $cH(z)/2\pi$ is 1.8 times larger at $z=1$. Weak lensing from Euclid- and Rubin-era surveys can test this.
2. The scale depends on ρ_Λ and not on H_0, so it should not track the Hubble-constant tension.
3. If dark energy evolves, $g_\dagger\propto\sqrt{\rho_{\mathrm{DE}}(z)}$, linking galaxy dynamics to the dark-energy history.

This proposal is developed within a broader reading of renormalisation scale as a coordinate [14], but the derivation and tests above do not depend on it.

Data and code availability

All data are public. Scripts reproducing every number above, with the pass rules written in each file, are available at `https://github.com/THEREALMIKEONE/a0-derivation-mike-one-repository`. The SPARC likelihood and the high-redshift test reuse the test suite of Ref. [14]. The KiDS-1000 and galaxy–galaxy lensing data were taken from the public compilation of R. Monjo [15].

Declarations

Funding. None. **Competing interests.** None. **Author contributions.** Mike One is the sole author.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used AI-assisted tools to write analysis code and to draft and edit the text. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

References

[1] S. S. McGaugh, F. Lelli and J. M. Schombert, Radial acceleration relation in rotationally supported galaxies, Phys. Rev. Lett. **117**, 201101 (2016). https://doi.org/10.1103/PhysRevLett.117.201101

[2] M. Milgrom, A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis, Astrophys. J. **270**, 365 (1983). https://doi.org/10.1086/161130

[3] M. Milgrom, The modified dynamics as a vacuum effect, Phys. Lett. A **253**, 273 (1999). https://doi.org/10.1016/S0375-9601(99)00077-8

[4] Planck Collaboration, Planck 2018 results. VI. Cosmological parameters, Astron. Astrophys. **641**, A6 (2020). https://doi.org/10.1051/0004-6361/201833910

[5] G. W. Gibbons and S. W. Hawking, Cosmological event horizons, thermodynamics, and particle creation, Phys. Rev. D **15**, 2738 (1977). https://doi.org/10.1103/PhysRevD.15.2738

[6] M. Milgrom, The MOND limit from space-time scale invariance, Astrophys. J. **698**, 1630 (2009). https://doi.org/10.1088/0004-637X/698/2/1630

[7] F. Lelli, S. S. McGaugh and J. M. Schombert, SPARC: mass models for 175 disk galaxies with Spitzer photometry at 3.6 μm and accurate rotation curves, Astron. J. **152**, 157 (2016). https://doi.org/10.3847/0004-6256/152/6/157

[8] T. Mistele, S. McGaugh, F. Lelli, J. Schombert and P. Li, Indefinitely flat circular velocities and the baryonic Tully–Fisher relation from weak lensing, Astrophys. J. Lett. **969**, L3 (2024).

[9] M. M. Brouwer et al., The weak lensing radial acceleration relation: constraining modified gravity and cold dark matter theories with KiDS-1000, Astron. Astrophys. **650**, A113 (2021). https://doi.org/10.1051/0004-6361/202040108

[10] S. H. Price et al., Rotation curves in z $\sim$ 1–2 star-forming disks, Astrophys. J. **922**, 143 (2021).

[11] J. P. Stott et al., The KMOS Redshift One Spectroscopic Survey (KROSS), Mon. Not. R. Astron. Soc. **457**, 1888 (2016). https://doi.org/10.1093/mnras/stw129

[12] E. Wisnioski et al., The KMOS^{3D} survey: data release and final survey paper, Astrophys. J. **886**, 124 (2019).

[13] A. L. Tiley et al., The KMOS Galaxy Evolution Survey (KGES), Mon. Not. R. Astron. Soc. **506**, 323 (2021).

[14] M. One, A scale-dependent vacuum: vacuum energy along the renormalisation scale, one acceleration scale from galaxies to clusters, and a fading dark energy (2026), https://github.com/THEREALMIKEONE/A-scale-dependent-vacuum-Repository.

[15] R. Monjo, lensing_RAR_KiDS, public data and code repository, https://github.com/robertmonjo/lensing_RAR_KiDS.